Juan Carlos Vega Rojano

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Systems Engineering student (10th semester) focused on Network Operations (NOC L1) and IT Infrastructure Support. Solid knowledge of TCP/IP models, IP addressing, VLSM, VLAN segmentation, inter-VLAN routing, traffic analysis, and Layer 2/3 security. Hands-on experience monitoring infrastructure with Zabbix and building network scanning tools with Python, using a methodical approach to technical incident resolution. Internationally certified bilingual profile (EF SET B2 - Upper Intermediate). Demonstrated ability to work in teams, attention to detail, and assertive communication.

PROJECTS

Zabbix Monitoring Lab [View on GitHub]
Infrastructure and Monitoring Administrator — Independent Project (Home Lab) *August 2026*

- Deployed a monitoring lab with Zabbix Server, Frontend, and Agent on an Ubuntu Server 24.04 LTS VM (VirtualBox), with hardened MariaDB and segmented network (NAT + Host-Only), managed via SSH.
- Configured ICMP availability checks and "host down" triggers across two hosts, validating the full cycle with a controlled failure test (trigger and auto-resolution with no manual intervention).
- Built a NOC-style dashboard and pruned about 60 irrelevant auto-discovered triggers to reduce false alarms, diagnosing and resolving real incidents by isolating the affected layer (virtualization, network, application, database).

TechNova VLAN Lab [View on GitHub]
Network Designer — Independent Project (Cisco Packet Tracer) *July 2026*

- Designed and implemented a multi-VLAN corporate network in Cisco Packet Tracer applying VLSM subnetting (192.168.10.0/24 block) to optimize IP addressing.
- Configured router-on-a-stick inter-VLAN routing (802.1Q), achieving 100% connectivity and interconnection across authorized traffic flows.
- Hardened Layer 2 and Layer 3 security with Port Security and extended ACLs, integrating DHCP and DNS services per VLAN and validating the mitigation of unauthorized access.

Network Analyzer Dashboard [View on GitHub]
Infrastructure and Network Developer — Independent Open Source Project *May 2026 – June 2026*

- Developed a Python network analysis and scanning tool that combines ARP table reading and multithreaded ICMP sweeps to detect active nodes on the local network.
- Programmed an automated module for VLSM subnet calculation and optimization, reducing addressing planning time.
- Designed an HTML control panel (dashboard) with an interactive network topology to diagnose device connectivity and availability in real time.

EDUCATION

UNIVERSIDAD SIMÓN BOLÍVAR **Barranquilla, Colombia**
Undergraduate Degree in Systems Engineering (10th Semester — In Progress) *Feb 2022 - Present*

CERTIFICATIONS

EF SET English Certificate (B2 Upper Intermediate) [View Credential]
EF Standard English Test *July 2026*

Networking Basics [View Credential]
Cisco Networking Academy *June 2026*

Administrador de Bases de Datos [View Credential]
Capacítate para el Empleo *September 2025*

ADDITIONAL SKILLS

- **Networking & Infrastructure:** TCP/IP, Cisco Packet Tracer, VLANs, Inter-VLAN Routing (802.1Q), ACLs, Port Security, IP Addressing / VLSM, DHCP, DNS.
- **Network Monitoring & Analysis:** Zabbix (Server, Frontend, Agent), NOC trigger and dashboard design, ARP/ICMP scanning with Python, connectivity diagnostics (Troubleshooting).
- **Operating Systems & Virtualization:** Linux / Ubuntu Server (remote administration via SSH), Oracle VirtualBox, MariaDB.
- **Databases:** MySQL, MariaDB, Oracle SQL.
- **Languages:** Spanish (Native), English (Upper Intermediate - B2 Certified).